vmspawnd — Enterprise VM Management Platform

The Problem

Organizations running Linux infrastructure need a unified way to manage virtual machines. Existing solutions are either:

- **Too heavy** — VMware vSphere, Proxmox, and OpenStack require complex multi-server deployments, dedicated storage infrastructure, and specialized operations teams
- **Too basic** — Manual QEMU/KVM management with shell scripts doesn't scale and lacks security, monitoring, or multi-user access
- **Too locked-in** — Cloud-only solutions (AWS, Azure, GCP) create vendor dependency with unpredictable costs

vmspawnd fills the gap — a single-binary VM management platform that runs on any Linux server with systemd, providing enterprise features without enterprise complexity.

What Is vmspawnd?

vmspawnd is a production-grade virtual machine management platform built in Rust. It wraps systemd-vmspawn and systemd-machined with a complete management layer:

- **One binary, one config file, one systemd service** — deploys in under 5 minutes
- **520+ REST API endpoints** with JWT authentication, RBAC, and audit logging
- **5 management interfaces** — CLI, terminal UI, web dashboard, Kubernetes operator, Terraform provider
- **Enterprise features** — HA clustering, live migration, GPU passthrough, backup/restore, network policies

```
sudo systemctl enable --now vmspawnd
# That's it. Platform is running.
```

Key Differentiators

1. Built on systemd (Not a Custom Hypervisor)

vmspawnd leverages systemd-vmspawn and systemd-machined — the VM management tools built into every modern Linux distribution. This means:

- No custom kernel modules or hypervisor patches
- VMs are first-class systemd units with journal logging, socket activation, and watchdog support
- Works on Fedora, Ubuntu, Debian, RHEL, SUSE — any distro with systemd 256+
- Upstream-maintained VM lifecycle, not a fork

2. Single Binary, Zero Dependencies

vmspawnd	Proxmox	OpenStack
1 binary (15MB)	200+ packages	1000+ packages
1 config file	50+ config files	100+ config files
5 min setup	2+ hours	2+ days
Runs on any Linux	Debian only	Ubuntu/RHEL
SQLite user store	PostgreSQL required	MySQL + RabbitMQ + Memcached

3. Security-First Architecture

The entire codebase has undergone a **21-round security audit** with 161 issues identified and fixed:

- **Zero unsafe Rust** — memory-safe by construction
- **Zero shell pipelines** — all subprocess calls use safe argument passing
- **JWT + RBAC** on every API endpoint (Admin/User/Viewer roles)
- **bcrypt password hashing** with auto-generated secrets (0600 file permissions)
- **Rate limiting** on authentication (5 attempts/5 min)
- **Input validation** on every user-facing parameter
- **Audit logging** on all VM lifecycle operations
- **Path traversal protection** with canonicalization
- **SQL injection prevention** — all queries parameterized

4. Rust Performance

- **Sub-millisecond API response times** — async Axum + Tokio runtime
- **Low memory footprint** — ~20MB RSS for the daemon
- **No garbage collection pauses** — predictable latency
- **Safe concurrency** — per-VM mutexes prevent race conditions

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Feature Overview

VM Lifecycle

- Create, start, stop, restart, pause, resume, delete
- Full and linked cloning with CoW support
- **Hibernate (suspend-to-disk)** and resume from snapshot
- Templates for rapid deployment
- Declarative config via YAML (`vmctl apply -f config.yaml`)
- Multiple disk formats: qcow2, raw, vmdk, vdi
- **VM import** from VMDK, VDI, VHD (auto-convert to qcow2)
- **Online disk resize** (qemu-img + QMP block_resize)

Storage

- **6 backends**: Local, NFS, LVM, LVM-thin, ZFS, Ceph/RBD
- Volume CRUD with attach/detach, resize, clone
- Snapshot create/restore with retention policies
- ZFS replication with incremental send/receive
- Ceph cluster health monitoring and RBD image management
- **Storage live migration** — move VM disks between pools without downtime
- **Cloud image downloader** — built-in catalog (Ubuntu, Fedora, Debian, Alma)
- **ISO management** — download, list, delete ISO images

Networking

- **Network Policies** — Cilium-style label-based ingress/egress rules
- **VM Firewall** — Per-VM firewall profiles and zones via nftables
- **Service Mesh** — Virtual IP load balancing (round-robin, least-conn, random, IP-hash)
- **QoS / Traffic Shaping** — Guaranteed/max rate, burst, priority-based bandwidth

- **DNS Policy** — Zone management, upstream servers, domain blocking
- **VPN Mesh** — WireGuard tunnels (point-to-point, hub-spoke, full-mesh)
- **Packet Mirror** — Traffic capture for debugging
- **NAT Gateway** — Masquerade, SNAT, DNAT, hairpin NAT
- **Network Monitor** — Per-VM bandwidth tracking with threshold alerts

Security & Identity

- JWT authentication with configurable token expiration
- **LDAP and OIDC/OAuth2** integration for enterprise SSO
- 3-tier RBAC (Admin / User / Viewer) enforced on every endpoint
- **Multi-tenancy** — project isolation with member roles and quotas
- API keys for service-to-service authentication
- TLS/HTTPS with certificate management
- Audit logging with JSON/CSV export
- Encryption at rest
- Rate limiting on authentication and API keys
- **21-round security audit** — 161 issues fixed, zero outstanding vulnerabilities
- **Storage pool name validation** — LVM, ZFS, Ceph pool names validated
- **SSRF prevention** on all user-provided URLs
- **Entity ID sanitization** in state store
- **Credential redaction** in API responses
- **WebSocket authentication** on console and VNC endpoints

High Availability

- etcd-based clustering with leader election
- Predictive DRS (Distributed Resource Scheduling)
- **Affinity / anti-affinity rules** for VM placement constraints
- Fault tolerance with automatic failover and fencing
- Distributed storage replication
- Site recovery with failover/reprotect workflows
- **Resource overcommit policies** (CPU/memory/storage ratios)

Monitoring & Automation

- Prometheus metrics exporter (`/metrics` endpoint)
- **Metrics retention policies** with configurable cleanup
- Analytics dashboard with historical data
- Multi-channel notifications: Email, Slack, **Webhook with retry + backoff**, Microsoft Teams
- VM scheduling (once, daily, weekly)

- Backup/restore with retention policies and incremental backups
- Resource quotas, pools, and datacenter abstractions
- **Database schema migrations** with version tracking

Console Access

- WebSocket terminal via xterm.js (browser-based SSH)
- VNC graphical console via noVNC proxy
- Authenticated with same JWT tokens as API

Cloud & Virtualization

- cloud-init integration (NoCloud datasource)
- TPM/vTPM support (1.2 and 2.0 via swtpm)
- GPU passthrough (NVIDIA, AMD, Intel GVT-g)
- **Live migration** with iterative rsync pre-copy and cutover
- CPU pinning and NUMA optimization
- **IPv6 support** — dual-stack nftables (ip + ip6)
- **API versioning** — all endpoints under `/api/` and `/api/v1/`

Management Interfaces

CLI (`vmctl`)

Scriptable command-line tool with JSON/YAML/table output:

```
vmctl list -o json
vmctl create myvm --image=ubuntu.qcow2 --cpus=4 --memory=4G
vmctl start myvm
vmctl apply -f infrastructure.yaml
vmctl policy list
vmctl ceph health my-pool
```

Terminal UI (`vmctl-tui`)

k9s-style dashboard with 8 views, vim keybindings, sparkline graphs, and live API data.

Web Dashboard

React-based UI with 37+ pages, command palette (Ctrl+K), dark theme, real-time WebSocket updates, and bulk operations.

Kubernetes Operator

Manage VMs as `VirtualMachine` CRDs with automatic reconciliation:

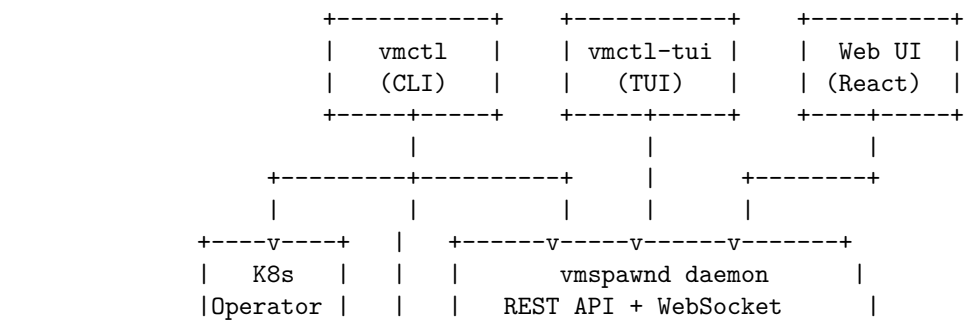
```yaml
apiVersion: vmspawnd.io/v1
kind: VirtualMachine
metadata:
  name: web-server
spec:
  image: ubuntu-22.04.qcow2
  cpus: 4
  memory: 4096
```

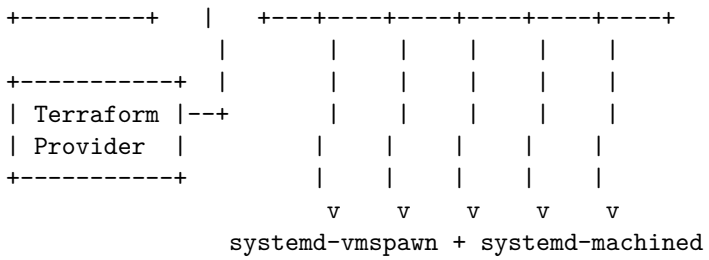
Terraform Provider

Declarative VM provisioning with full plan/apply workflow:

```
resource "vmspawnd_vm" "web" {
  name   = "web-server"
  image  = "ubuntu-22.04.qcow2"
  cpus   = 4
  memory = 4096
}
```

Architecture





Deployment Models

Single Server

One Linux server running vmspawnd with local storage. Suitable for development, testing, small teams, and edge deployments.

Requirements: Linux with systemd 256+, 4GB RAM minimum

Multi-Node Cluster

Multiple vmspawnd nodes with etcd-based clustering, shared storage (NFS/Ceph), live migration, and HA failover.

Requirements: 3+ nodes, shared storage, etcd cluster

Kubernetes-Managed

vmspawnd nodes managed by the Kubernetes operator. VMs defined as CRDs alongside containerized workloads.

Requirements: Kubernetes cluster with vmspawnd operator deployed

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Comparison

Feature	vmspawnd	Proxmox VE	OpenStack	libvirt/virsh
Single-binary deployment	Yes	No	No	N/A
REST API	520+ endpoints	~50	~200	XML-RPC
Web UI	Yes	Yes	Yes (Horizon)	No

Feature	vmspawnd	Proxmox VE	OpenStack	libvirt/virsh
CLI	Yes	Yes	Yes	Yes
Terminal UI	Yes	No	No	No
Kubernetes Operator	Yes	No	Yes	No
Terraform Provider	Yes	Yes	Yes	Yes
Network Policies	Cilium-style	Basic	Neutron	No
Service Mesh	Yes	No	No	No
VPN Mesh	WireGuard	No	No	No
GPU Passthrough	Yes	Yes	Yes	Yes
Live Migration	Yes	Yes	Yes	Yes
LDAP/OIDC SSO	Yes	Yes	Yes (Keystone)	No
Multi-tenancy	Yes	Yes	Yes	No
RBAC	3-tier	3-tier	Keystone	No
VM Hibernate	Yes	Yes	No	Yes
Storage Live Migration	Yes	Yes	Yes	Yes
VM Import (VMDK/VDI)	Yes	Yes	Limited	qemu-img
Audit Logging	Yes	Yes	Yes	No
Written in	Rust	Perl/C	Python	C
Memory Safety	Guaranteed	No	N/A	No
Setup Time	5 min	2 hours	2 days	Manual
License	MIT	AGPL	Apache 2.0	LGPL

Technology Stack

Layer	Technology
Language	Rust (2021 edition)
Async Runtime	Tokio 1.44
Web Framework	Axum 0.8
Terminal UI	ratatui + crossterm
Web UI	React 18 + TypeScript + Vite + TailwindCSS
VM Backend	systemd-vmspawn + systemd-machined
D-Bus	zbus 4
Monitoring	Prometheus

Project Statistics

Metric	Value
Backend crates	40
Rust source files	165
TypeScript source files	130
Total lines of code	~87,000
REST API endpoints	520+
WebSocket endpoints	3
Web pages	37+
Security audit rounds	21
Security issues fixed	161
Test suite	Passing

License

MIT — free for commercial use, modification, and distribution.

Getting Started

```
# Build
cd backend && cargo build --release

# Install
sudo make install

# Start
sudo systemctl enable --now vmspawnd

# Read admin password
sudo cat /var/lib/vmspawnd/.admin_password

# Login
curl -X POST http://localhost:8080/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"username": "admin", "password": "<password>"}'

# Create your first VM
vmctl create myvm --image=/path/to/image.qcow2 --cpus=4 --memory=4096
vmctl start myvm
```

```
# Open web dashboard
open http://localhost:8080
```

For technical details, see the Architecture Guide, API Reference, and Security Documentation.